Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • Grade: 85.71%

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Projection Onto General Subspaces

0:10 / 1:13 2x

For a subspace with multiple basis vectors $b_1, b_2, \ldots, b_m$, the projection matrix is $P = B(B^TB)^{-1}B^T$, where B is the matrix of basis vectors.

This projection retains maximum information from the original data in the reduced dimension, crucial for techniques like Principal Component Analysis (PCA).

Figure 3.11 from *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, demonstrate the projection of a vector x onto a two-dimensional subspace U in $\mathbb{R}^3$. The figure shows a vector $x$and its orthogonal projection $\pi_U(x)$ onto the plane defined by the basis vectors b_1 and b_2. The projection $\pi_U(x)$ is depicted as the sum of the components of x along b_1 and b_2. The vector $x-\pi_U(x)$, which is orthogonal to the subspace U, represents the error vector and is perpendicular to the plane. This figure illustrates the concept of projecting a vector onto a higher-dimensional subspace, where the projection matrix P is used to find $\pi_U(x)$, ensuring that $x-\pi_U(x)$ lies entirely in the orthogonal complement of U.

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